

E commerce Growth Strategy & Operations Analytics

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Tools Used :- Advance Excel, Excel Charts, KPI Dashboarding, SQL Server

Project Overview

- The Objective of this project is to analyse business performance and identify opportunities for revenue growth , customer retention and operational improvement
- The project focuses on answering key business questions related to customers behaviour, product performance , revenue trends, and operational efficiency
- Through data analysis and interactive dashboarding , actionable insights and business recommendations were developed to support strategic decision making and business growth
- This project demonstrates skills in SQL querying , data modelling , data transformation, dashboard development , KPI reporting and business analysis

Problem Statement

An e-commerce company is experiencing slowing revenue growth , declining repeat purchase , and varying levels of operational performance .

It requires a data driven approach to identify key factors affecting the revenue , customer retention, product performance and delivery efficiency in order to support business growth and improve decision making

Project Objectives

- Analyse revenue trends and growth drivers
- Identify high value customers and customer retention patterns
- Evaluate product and category performance
- Measure operational efficiency through order and delivery analysis
- Generate actionable business recommendations based on data insights
- Develop an interactive reporting solution using SQL , excel, power query and power pivot

Dataset Overview

Dataset Name

Olist Brazilian E-Commerce Dataset

Dataset Description

The dataset contains transactional data from a Brazilian e-commerce marketplace. It includes information related to customers, orders, products, sellers, payments, and reviews. The dataset is used to analyze business performance, customer behavior, revenue trends, and operational efficiency.

Tables Used

- Customers – 99,441 records
- Orders – 99,441 records
- Order Items – 112,650 records
- Order Payments – 103,886 records
- Order Reviews – 99,224 records
- Products – 32,951 records
- Sellers – 3,095 records

Project Workflow

Olist Dataset



Data Understanding



SQL Data Analysis



Business Question Analysis



Excel Data Preparation



Dashboard Development



Business Insights



Recommendations



Conclusion

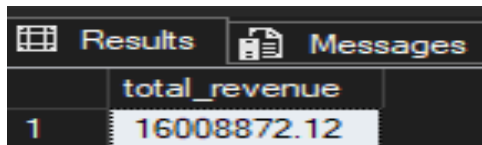
Revenue Analysis

1. What is the total revenue generated by the company ?

SQL Query:

```
SELECT SUM(payment_value) AS total_revenue FROM  
order_payments
```

Result :



A screenshot of a SQL query results window. It has two tabs: 'Results' and 'Messages'. The 'Results' tab is active, showing a single row with two columns: 'total_revenue' and '16008872.12'.

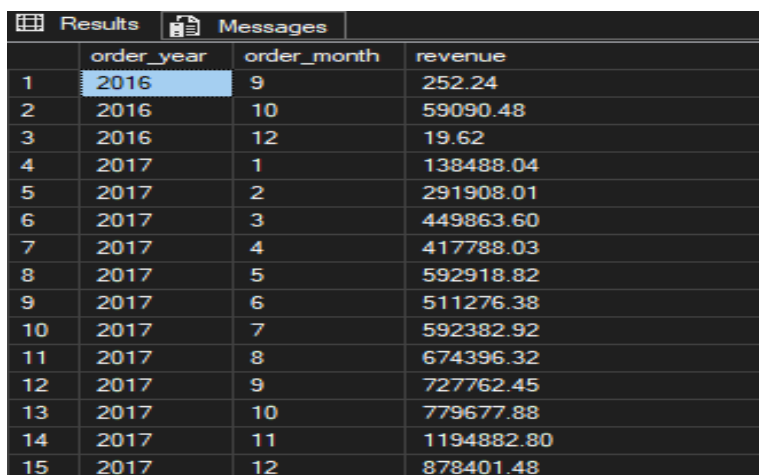
	total_revenue
1	16008872.12

2. How has monthly revenue changed over time?

SQL Query:

```
SELECT YEAR (o.order_purchase_timestamp) as order_year,  
MONTH (o.order_purchase_timestamp) as order_month,  
ROUND(SUM(op.payment_value),2) as revenue  
FROM orders o  
JOIN order_payments op ON o.order_id = op.order_id  
GROUP BY YEAR (o.order_purchase_timestamp) ,MONTH  
(o.order_purchase_timestamp)  
ORDER BY order_year, order_month
```

Result :



A screenshot of a SQL query results window. It has two tabs: 'Results' and 'Messages'. The 'Results' tab is active, showing a table with four columns: 'order_year', 'order_month', and 'revenue'. The table contains 15 rows of data, starting from 2016 and ending in 2017.

	order_year	order_month	revenue
1	2016	9	252.24
2	2016	10	59090.48
3	2016	12	19.62
4	2017	1	138488.04
5	2017	2	291908.01
6	2017	3	449863.60
7	2017	4	417788.03
8	2017	5	592918.82
9	2017	6	511276.38
10	2017	7	592382.92
11	2017	8	674396.32
12	2017	9	727762.45
13	2017	10	779677.88
14	2017	11	1194882.80
15	2017	12	878401.48

3. Which product categories generate the highest revenue ?

SQL Query:

```
SELECT TOP 5 p.product_category_name,  
ROUND(SUM(OI.price),2) as product_revenue  
FROM order_items oi  
JOIN products p ON oi.product_id = p.product_id  
GROUP BY p.product_category_name
```

Result :

	product_category_name	product_revenue
1	dvds_blu_ray	5999.39
2	artes	24202.64
3	telefonica	323667.53
4	fraldas_higiene	1567.59
5	industria_comercio_e_negocios	39669.61

4. Rank product categories by revenue contribution ?

SQL Query:

```
SELECT p.product_category_name,  
ROUND(SUM(oi.price),2) as product_revenue,  
DENSE_RANK() OVER(order by sum(oi.price) DESC) as  
revenue_rank  
FROM order_items oi  
JOIN products p ON oi.product_id = p.product_id  
GROUP BY p.product_category_name  
ORDER BY revenue_rank
```

Result :

	product_category_name	product_revenue	revenue_rank
1	beleza_saude	1258681.34	1
2	relogios_presentes	1205005.68	2
3	cama_mesa_banho	1036988.68	3
4	esporte_lazer	988048.97	4
5	informatica_acessorios	911954.32	5
6	moveis_decoracao	729762.49	6
7	cool_stuff	635290.85	7
8	utilidades_domesticas	632248.66	8
9	automotivo	592720.11	9
10	ferramentas_jardim	485256.46	10

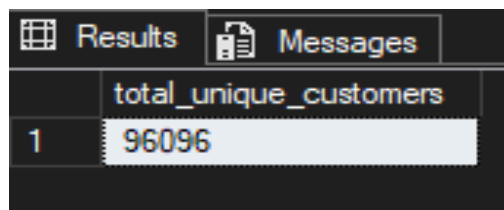
Customer Analysis

5. How many unique customers does the company have ?

SQL Query:

```
SELECT COUNT(distinct customer_unique_id)
as total_unique_customers FROM customers
```

Result :



Results		Messages	
		total_unique_customers	
1		96096	

6. Top 10 customers by revenue ?

SQL Query:

```
SELECT TOP 10 c.customer_unique_id,
SUM(op.payment_value) as revenue, DENSE_RANK () OVER(order
by SUM(op.payment_value) DESC) as customer_rank FROM
customers c JOIN orders o ON c.customer_id = o.customer_id
JOIN order_payments op ON o.order_id = op.order_id GROUP
BY c.customer_unique_id ORDER BY revenue DESC
```

Result :

	customer_unique_id	revenue	customer_rank
1	0a0a92112bd4c708ca5fde585afaa872	13664.08	1
2	46450c74a0d8c5ca9395da1daac6c120	9553.02	2
3	da122df9eeddfedc1dc1f5349a1a690c	7571.63	3
4	763c8b1c9c68a0229c42c9fc6f662b93	7274.88	4
5	dc4802a71eae9be1dd28f5d788ceb526	6929.31	5
6	459bef486812aa25204be022145caa62	6922.21	6
7	ff4159b92c40ebe40454e3e6a7c35ed6	6726.66	7
8	4007669dec559734d6f53e029e360987	6081.54	8
9	5d0a2980b292d049061542014e8960bf	4809.44	9
10	eebb5dda148d3893cdaf5b5ca3040ccb	4764.34	10

7. Find out total number of repeat customers ?

SQL Query:

```
WITH customers_orders AS
(SELECT c.customer_unique_id, COUNT(o.order_id) AS
total_orders
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_unique_id )
SELECT COUNT(*) as repeat_customers
FROM customers_orders
WHERE total_orders > 1
```

Result :

Results		Messages	
repeat_customers			
1	2997		

Product Analysis

8. Which product categories have highest sales volume ?

SQL Query:

```
SELECT product_category_name ,COUNT (*) AS units_sold
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
GROUP BY p.product_category_name
ORDER BY units_sold DESC
```

Result:

	product_category_name	units_sold
1	cama_mesa_banho	11115
2	beleza_saude	9670
3	esporte_lazer	8641
4	moveis_decoracao	8334
5	informatica_acessorios	7827
6	utilidades_domesticas	6964
7	relogios_presentes	5991
8	telefonica	4545
9	ferramentas_jardim	4347
10	automotivo	4235

9. Which product categories generate above average revenue ?

SQL Query:

```
WITH category_revenue AS (  
SELECT p.product_category_name, SUM (oi.price) as revenue  
FROM products p  
JOIN order_items oi ON p.product_id = oi.product_id  
GROUP BY p.product_category_name)
```

```
SELECT * FROM category_revenue  
WHERE revenue > (SELECT AV(revenue) FROM  
category_revenue)  
ORDER BY revenue DESC
```

Result:

	product_category_name	revenue
1	beleza_saude	1258681.34
2	relogios_presentes	1205005.68
3	cama_mesa_banho	1036988.68
4	esporte_lazer	988048.97
5	informatica_acessorios	911954.32
6	moveis_decoracao	729762.49
7	cool_stuff	635290.85
8	utilidades_domesticas	632248.66
9	automotivo	592720.11
10	ferramentas_jardim	485256.46

10. Which product categories have the highest average customer ratings?

SQL Query:

```
SELECT p.product_category_name,  
ROUND(AV(CAST (r.review_score AS FLOAT )),2) AS  
avg_rating,  
COUNT(r.review_id) AS total_review  
FROM products p  
JOIN order_items oi ON p.product_id = oi.product_id  
JOIN order_reviews r ON oi.order_id = r.order_id  
GROUP BY p.product_category_name  
HAVING COUNT(r.review_id) >=100  
ORDER BY avg_rating DESC
```

Result:

Results		Messages	
	product_category_name	avg_rating	total_review
1	livros_interesse_geral	4.45	549
2	livros_tecnicos	4.37	266
3	malas_acessorios	4.32	1088
4	alimentos_bebidas	4.32	279
5	fashion_calcados	4.23	261
6	alimentos	4.22	495
7	pet_shop	4.19	1939
8	papelaria	4.19	2507
9	pcs	4.17	200
10	eletrodomesticos	4.17	806
11	brinquedos	4.16	4091

Operations Analysis

11. Which seller generate the highest revenue and how do they rank ?

SQL Query:

```
SELECT s.seller_id, SUM (oi.price) as revenue,  
DENSE_RANK() OVER(order by SUM (oi.price )DESC) as ranking  
FROM sellers s  
JOIN order_items oi ON s.seller_id = oi.seller_id  
JOIN order_payments op ON oi.order_id = op.order_id  
GROUP BY s.seller_id  
ORDER BY revenue DESC
```

Result:

	seller_id	revenue	ranking
1	53243585a1d6dc2643021fd1853d8905	244627.55	1
2	4869f7a5dfa277a7dca6462dcf3b52b2	237867.23	2
3	4a3ca9315b744ce9f8e9374361493884	213299.57	3
4	fa1c13f2614d7b5c4749cbc52fecda94	203984.22	4
5	7c67e1448b00f6e969d365cea6b010ab	198194.33	5
6	7e93a43ef30c4f03f38b393420bc753a	182878.17	6
7	da8622b14eb17ae2831f4ac5b9dab84a	169297.77	7
8	7a67c85e85bb2ce8582c35f2203ad736	149939.84	8
9	1025f0e2d44d7041d6cf58b6550e0bfa	142130.94	9
10	955fee9216a65b617aa5c0531780ce60	137335.00	10

12. What percentage of orders were delivered late ?

SQL Query:

```
SELECT ROUND(100 * SUM(CASE WHEN  
order_delivered_customer_date >  
order_estimated_delivery_date THEN 1 ELSE 0  
END) / COUNT(*),2 ) AS late_delivery_percentage FROM  
orders WHERE order_delivered_customer_date is NOT NULL
```

Result:



Results		Messages	
	late_delivery_percentage		
1	8		

13. Do delivery delays impact customer review scores?

SQL Query:

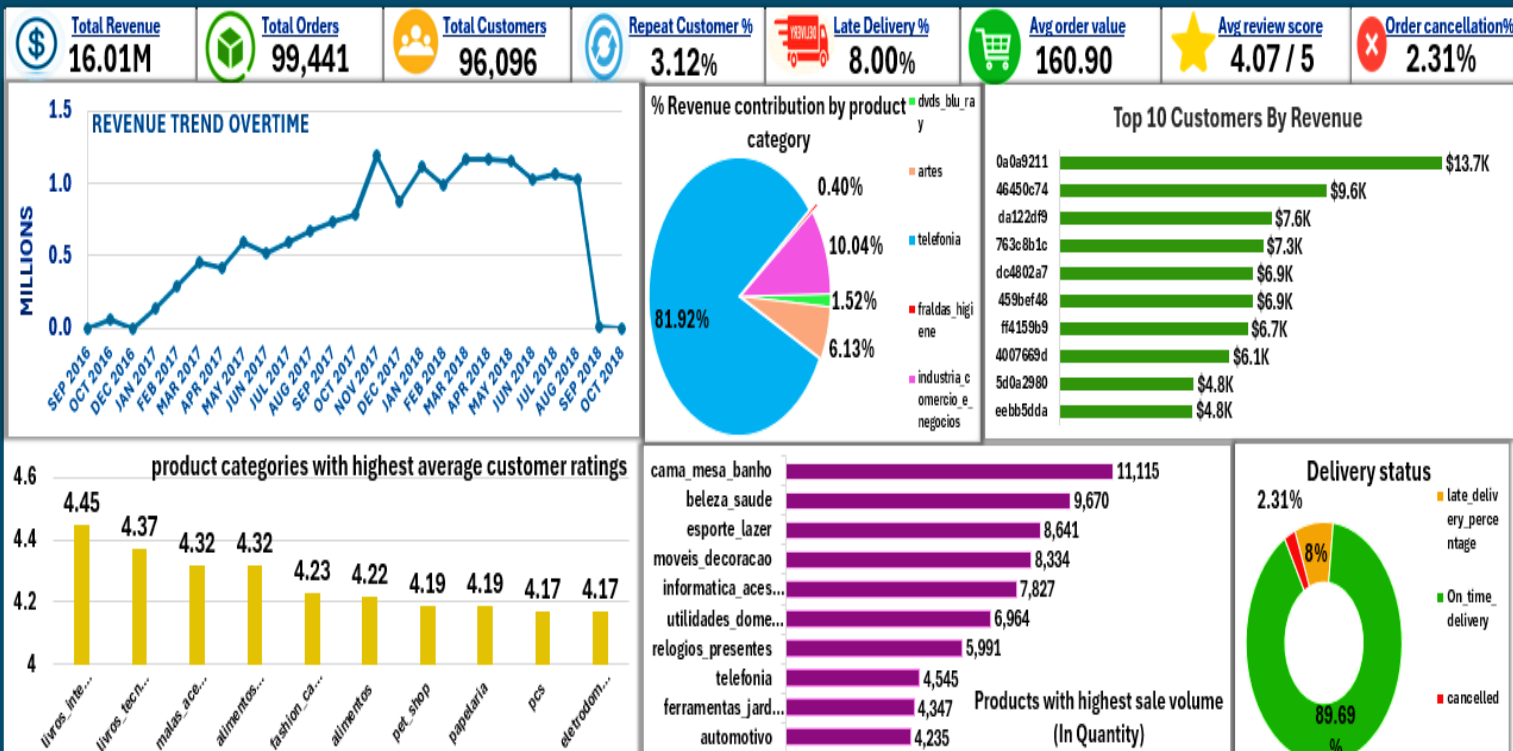
```
SELECT CASE
WHEN o.order_delivered_customer_date >
o.order_estimated_delivery_date
THEN 'Late Delivery'
ELSE 'On Time'
END AS Delivery_Status,
ROUND(AV(CAST(r.review_score AS FLOAT)),2) AS
Average_Review_Score, COUNT(*) AS Total_Orders
FROM orders o
JOIN order_reviews r ON o.order_id = r.order_id
WHERE o.order_delivered_customer_date IS NOT NULL
GROUP BY
CASE
WHEN o.order_delivered_customer_date >
o.order_estimated_delivery_date
THEN 'Late Delivery'
ELSE 'On Time'
END
```

Result:

	Delivery_Status	Average_Review_Score	Total_Orders
1	On Time	4.29	88658
2	Late Delivery	2.57	7701

E-Commerce Business Performance Dashboard

E-COMMERCE BUSINESS PERFORMANCE DASHBOARD



Dashboard Highlights

- Revenue and order KPIs provide a quick overview of business performance.
- Monthly trend identifies revenue growth patterns.
- Customer analysis identifies high-value customers.
- Product analysis highlights top-selling categories.
- Delivery performance supports operational monitoring.

Key Insights

1. Revenue growth followed by a slowdown

- Monthly revenue showed a strong upward trend from late 2016 to early 2018, indicating sustained business growth before stabilizing toward the end of the analysis period

2. Revenue concentration

- A small number of product categories contributed a significant portion of total revenue, highlighting the importance of focusing inventory, marketing, and pricing strategies on these high-performing categories.

3. Customer retention opportunity

- Despite acquiring 96,096 customers, the repeat customer rate was only 3.12%, suggesting a significant opportunity to improve customer retention through loyalty programs and personalized marketing.

4. Delivery performance is good but has room for improvement

- Approximately 92% of orders were delivered on time, while 8% experienced delays. Reducing late deliveries could further enhance customer satisfaction and encourage repeat purchases.

5. Customer satisfaction remains strong

- The average customer review score of 4.07/5 indicates a generally positive customer experience, demonstrating that product quality and service levels are well received by customers.
- The top-performing customers generated substantially higher revenue than the average customer, emphasizing the value of identifying and retaining high-value customer segments through targeted engagement strategies.

Business Recommendation

- **Improve customer retention** by introducing loyalty programs, personalized promotions, and post-purchase engagement campaigns to increase the repeat customer rate.
- **Reduce late deliveries** by optimizing logistics operations, improving warehouse coordination, and monitoring delivery partners to improve on-time delivery performance.
- **Increase investment in top-performing product categories** through targeted marketing, better inventory planning, and promotional campaigns to maximize revenue growth.
- **Improve the performance of low-selling product categories** by reviewing pricing strategies, product assortment, customer demand, and promotional effectiveness.
- **Strengthen customer satisfaction initiatives** by analyzing negative reviews, addressing common customer concerns, and improving service quality to maintain and increase review scores.
- **Develop targeted marketing campaigns** for high-value customers by offering exclusive discounts, early product access, or loyalty rewards to increase customer lifetime value.
- **Monitor key business KPIs regularly** through an interactive dashboard to support data-driven decision-making and quickly identify performance trends and operational issues.

Conclusion

- Successfully analyzed the Olist E-Commerce dataset using Microsoft SQL Server and Microsoft Excel to evaluate business performance across revenue, customers, products, and operations.
- Applied advanced SQL concepts including JOINS, Common Table Expressions (CTEs), Window Functions, Aggregate Functions, Subqueries, CASE Statements, and GROUP BY to extract meaningful business insights from raw transactional data.
- Developed an interactive business dashboard in Excel featuring key performance indicators (KPIs) and visualizations to monitor revenue trends, customer behavior, product performance, delivery efficiency, and customer satisfaction.
- Identified that business revenue is concentrated in a limited number of high-performing product categories, highlighting opportunities for focused marketing and inventory optimization.
- Found that the repeat customer rate is relatively low, indicating significant potential to improve customer retention through loyalty programs and personalized engagement strategies.
- Customer satisfaction remained positive with an average review score above 4 out of 5, demonstrating strong product quality and service performance while identifying opportunities for continuous improvement.
- This project demonstrates the ability to transform raw business data into actionable insights, communicate findings effectively through data visualization, and support strategic business decision-making using data analytics.

Overall, the project highlights practical skills in SQL, Excel, KPI ,data-driven decision-making, making it a comprehensive end-to-end business analytics case study.